

No Adjoint-Induced Almost-Complex Structure on a Proper Unital Algebra Quotient

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Claimed status: full negative solution of the sole open question in Table 2 of arXiv:2410.22055, subject to human review. For the non-fixed crossed-product case, the table's question mark in the adjoint column is a minus sign. In fact the obstruction holds for every proper ideal of every unital associative algebra.

1 Source question

Let $\mathcal{A}$ be a unital C^* -algebra, let $\mathfrak{k}$ be the proper closed ideal used in the paper, and let G/K be the associated homogeneous space. Its tangent space at the base point is naturally $\mathcal{A}/\mathfrak{k}$. An admissible operator J induces an almost-complex structure only if

$$\text{Ran}(J^2 + I_{\mathcal{A}}) \subseteq \mathfrak{k}; \tag{1}$$

this is Definition 2.5 of the source. Equivalently, the operator induced by J on $\mathcal{A}/\mathfrak{k}$ must square to minus the identity.

The source considers $J = \text{ad}_d$, where $\text{ad}_d(a) = [d, a]$. Its final table leaves open whether such an operator can induce an almost-complex structure when

$$\mathcal{A} = C(X) \rtimes_{\alpha} \mathbb{Z}$$

and the quotient corresponds to a non-fixed invariant set.

2 Answer

Theorem 1 (Unit obstruction). *Let $\mathcal{A}$ be a unital associative algebra over $\mathbb{R}$ or $\mathbb{C}$, let $\mathfrak{k} \subsetneq \mathcal{A}$ be a proper two-sided ideal, and let $d \in \mathcal{A}$. Then the inner derivation $J = \text{ad}_d$ preserves $\mathfrak{k}$, but it cannot satisfy*

$$\text{Ran}(J^2 + I_{\mathcal{A}}) \subseteq \mathfrak{k}.$$

Equivalently, the induced inner derivation on the nonzero unital quotient $\mathcal{A}/\mathfrak{k}$ cannot be an almost-complex structure.

Idea of the proof

An almost-complex structure has square $-I$, so it is automatically invertible. An inner derivation can never be invertible on a nonzero unital quotient: it annihilates the unit. In the source's range formulation, this is visible by testing the single element 1.

The second table is meant to visualize the possibility of finding an almost complex structure within the described classes. One open question still remains.

	Rank one	LR mult.	adjoint
bounded	-	+	-
continuous	-	+	-
Toeplitz	-	+	-
crossed (fixed)	-	+	-
crossed (non-fixed)	-	+	?

TABLE 2. Possibility of almost complex structures: in each case the fact that an almost-complex structure can be constructed using the considered class of Nijenhuis operators is marked by a +-sign, when no almost-complex structure can be constructed we use a --sign, and when the question is still open we use a ?-sign.

Figure 1: The sole remaining almost-complex question in Table 2 on PDF page 14 of arXiv:2410.22055.

Proof. Because $\mathfrak{k}$ is a two-sided ideal, $[d, \mathfrak{k}] \subseteq \mathfrak{k}$, so J induces

$$\bar{J} = \text{ad}_{d+\mathfrak{k}} : \mathcal{A}/\mathfrak{k} \longrightarrow \mathcal{A}/\mathfrak{k}.$$

The quotient is nonzero and unital because $\mathfrak{k}$ is proper. Its unit obeys

$$\bar{J}(1 + \mathfrak{k}) = [d + \mathfrak{k}, 1 + \mathfrak{k}] = 0.$$

Thus $\bar{J}$ has a nonzero kernel and cannot satisfy $\bar{J}^2 = -I_{\mathcal{A}/\mathfrak{k}}$. Directly in $\mathcal{A}$, the same contradiction is the one-line calculation

$$(J^2 + I_{\mathcal{A}})(1) = 1 \notin \mathfrak{k}.$$

Hence the required range inclusion fails. $\square$

Corollary 2 (Resolution of the crossed-product question). *Let $\mathcal{A} = C(X) \rtimes_{\alpha} \mathbb{Z}$ and let $\tilde{\mathfrak{k}}_Y$ be any proper ideal used in the source, including the ideal associated with a non-fixed invariant set Y . For no $d \in \mathcal{A}$ does ad_d induce an almost-complex structure on the corresponding homogeneous space G/K .*

Proof. Apply Theorem 1 with $\mathfrak{k} = \tilde{\mathfrak{k}}_Y$. $\square$

Remark 3 (Stronger derivation form). *The proof uses only that the operator annihilates the unit. Therefore any derivation $D : \mathcal{A} \rightarrow \mathcal{A}$ that preserves a proper ideal $\mathfrak{k}$ fails to induce an almost-complex structure on $\mathcal{A}/\mathfrak{k}$: every derivation of a unital real or complex algebra has $D(1) = 0$.*

3 Scope and checks

The Nijenhuis condition is irrelevant to this obstruction. Even if an adjoint operator happens to have vanishing Nijenhuis torsion modulo the ideal, it already fails the square condition required of an almost-complex structure. Consequently this result fills the question mark in Table 2; it does *not* classify the still-partially-open adjoint entries in Table 1.

The source makes the same non-existence conclusion for the Calkin algebra using Wintner's theorem on commutators. The unit calculation above is stronger for this purpose and works uniformly for the crossed-product case left open in the source.

A bounded novelty check found only arXiv version 1, dated 29 October 2024. The sole OpenAlex-indexed citing item was an unrelated chapter on the infinite-dimensional Siegel disc; no later explicit resolution of the table question was found. The proof is algebraic and self-contained.

References

- [1] T. Goliński, G. Larotonda, and A. B. Tumpach, *Nijenhuis operators on homogeneous spaces related to C^* -algebras*, arXiv:2410.22055v1 (2024); International Journal of Geometric Methods in Modern Physics (2025), 2540040.